Sherrington - KirKpatricK model

Consider a N-site lattice



Configuration

$\sigma = \{\sigma_i\}_{i=1,\dots,N} \quad , \quad \sigma_i \in \{-1, 1\}$

Hamiltonian:

$$\mathcal{H}[\sigma] = -\sum_{i=1}^{N-1} \sum_{j=i+1}^{N} J_{ij}\,\sigma_i\sigma_j - h \sum_{i=1}^{N} \sigma_i =$$

$$= -\sum_{i<j} J_{ij}\,\sigma_i\sigma_j - h \sum_{i=1}^{N} \sigma_i$$

h: external magnetic field

Spin-glass model

→ random interactions, couplings drawn from

$$P(J) = \sqrt{\frac{N}{2\pi}}\, e^{-\frac{NJ^2}{2}}$$

Note: in the paper

$$\mathcal{H}[\sigma] = -\underbrace{\sum_{\langle i,j\rangle}}_{\substack{\text{nearest}\\ \text{neighbours}}} J_{ij}\,\sigma_i\sigma_j - h \sum_{i} \sigma_i$$

this is the Edward-Anderson model

Randomicity of the interaction is called quenched disorder

Quenched → thermodynamic properties have to be computed at fixed instant of the disorder

Partition function:

$$Z_J = \sum_{\{\sigma\}} e^{-\beta \mathcal{H}[\sigma]} =$$

$$= \sum_{\{\sigma\}} e^{\beta \sum_{i<j} J_{ij} \sigma_i \sigma_j + \beta h \sum_i \sigma_i}$$

where

$$\sum_{\{\sigma\}} = \sum_{\sigma_1 = -1,1} \sum_{\sigma_2 = -1,1} \dots \sum_{\sigma_N = -1,1} = \prod_{i=1}^{N} \sum_{\sigma_i = -1,1}$$

Free energy:

$$f = \lim_{N \to \infty} f_J^{(N)} \quad , \quad f_J^{(N)} = -\frac{1}{\beta} \frac{\ln Z_J}{N}$$

Quenched average (average over the disorder):

$$\mathbb{E}[f_J^{(N)}] = -\frac{1}{\beta} \frac{\mathbb{E}[\ln Z_J]}{N}$$

$N \to \infty$ $\Rightarrow$ self-averaging property:

$$\mathbb{P}\left(f = \lim_{N \to \infty} f_J^{(N)} = \lim_{N \to \infty} \mathbb{E}[f_J^{(N)}] \right) = 1$$

Note that there are sequences of J's such that

$$f = \lim_{N \to \infty} f_J^{(N)} \neq \lim_{N \to \infty} \mathbb{E}[f_J^{(N)}]$$

but these have zero measure

Then

$$f = \lim_{N \to \infty} \mathbb{E}[f_J^{(N)}] = -\frac{1}{\beta} \lim_{N \to \infty} \frac{\mathbb{E}[\ln Z_J]}{N}$$

Replica trick:

$$Z_J^m = e^{m \ln Z_J}$$

$m \ll 1$ $\Rightarrow$ $Z_J^m \simeq 1 + m \ln Z_J$

hence

$$\lim_{m \to 0} \frac{Z_J^m - 1}{m} = \ln Z_J$$

Thus

$$\lim_{m \to 0} \frac{\mathbb{E}[z_j^m] - 1}{m} = \mathbb{E}[\ln z_j]$$

Using

$$\ln(1+\delta) \simeq \delta \quad , \quad \delta = \mathbb{E}[z_j^m] - 1$$

then

$$\lim_{m \to 0} \frac{\ln(\mathbb{E}[z_j^m])}{m} = \mathbb{E}[\ln z_j]$$

Note that $\ln(\mathbb{E}[z_j^m])$ is simpler to compute than $\mathbb{E}[\ln z_j]$

Then

$$f = \lim_{N \to \infty} \mathbb{E}[f_j^{(N)}] = -\frac{1}{\beta} \lim_{N \to \infty} \frac{\mathbb{E}[\ln z_j]}{N} =$$

$$= -\frac{1}{\beta} \lim_{N \to \infty} \lim_{m \to 0} \frac{\ln(\mathbb{E}[z_j^m])}{Nm}$$

Assumption: we can exchange the two limits $N \to \infty$, $m \to 0$

So

$$f = -\frac{1}{\beta} \lim_{m \to 0} \lim_{N \to \infty} \frac{\ln(\mathbb{E}[z_j^m])}{Nm}$$

We compute $\mathbb{E}[z_j^m]$ for m integer, then we take an analytic continuation.

$$z_j^m = \prod_{a=1}^{m} z_j =$$

$$= \sum_{\{\vec{\sigma}\}} e^{\beta \sum_{a=1}^{m} \sum_{i<j} J_{ij} \sigma_i^a \sigma_j^a + \beta h \sum_{a=1}^{m} \sum_i \sigma_i^a}$$

where now

$$\sum_{\{\vec{\sigma}\}} = \prod_{a=1}^{m} \prod_{i=1}^{N} \sum_{\sigma_i^a = -1, 1}$$

(m replicas of the system living in the same realization of the disorder)

$$\mathbb{E}\left[Z_J^m\right] = \int \left(\prod_{i<j} dJ_{ij}\, P(J_{ij})\right) \sum_{\{\vec{\sigma}\}} e^{\beta \sum_a \sum_{i<j} J_{ij} \sigma_i^a \sigma_j^a + \beta h \sum_a \sum_i \sigma_i^a} =$$

$$= \sum_{\{\vec{\sigma}\}} e^{\beta h \sum_a \sum_i \sigma_i^a} \prod_{i<j} \sqrt{\frac{N}{2\pi}} \int dJ_{ij}\, e^{-\frac{N J_{ij}^2}{2} + \beta \sum_a J_{ij} \sigma_i^a \sigma_j^a}$$

Using

$$\int dx\, e^{-\frac{1}{2}Ax^2 + Bx} = \sqrt{\frac{2\pi}{A}}\, e^{\frac{B^2}{2A}}$$

(gaussian integral)

with

$$A = N \quad , \quad B = \beta \sum_a \sigma_i^a \sigma_j^a$$

then

$$\mathbb{E}\left[Z_J^m\right] = \sum_{\{\vec{\sigma}\}} e^{\beta h \sum_a \sum_i \sigma_i^a} \prod_{i<j} e^{\frac{\beta^2}{2N} \left(\sum_a \sigma_i^a \sigma_j^a\right)^2}$$

$$= \sum_{\{\vec{\sigma}\}} e^{\beta h \sum_a \sum_i \sigma_i^a + \frac{\beta^2}{2N} \sum_{i<j} \sum_{a,b} \sigma_i^a \sigma_j^a \sigma_i^b \sigma_j^b}$$

We have introduced an "interaction" between replicas, namely replicas are no longer indipendent.

$$\sum_{i<j} = \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} = \frac{1}{2} \sum_{i=1}^{N} \sum_{\substack{j=1 \\ j\neq i}}^{N} = \frac{1}{2} \left(\sum_{i,j=1}^{N} - \sum_{i,j=1}^{N} \delta_{ij} \right) = \frac{1}{2} \sum_{i,j=1}^{N} (1 - \delta_{ij})$$

$$\sum_{i<j} \sum_{a,b} \sigma_i^a \sigma_j^a \sigma_i^b \sigma_j^b =$$

$$= \frac{1}{2} \sum_{i,j=1}^{N} \sum_{a,b=1}^{m} \sigma_i^a \sigma_i^b \sigma_j^a \sigma_j^b - \frac{1}{2} \sum_{i=1}^{N} \sum_{a,b=1}^{m} \underbrace{\sigma_i^a \sigma_i^a}_{1} \underbrace{\sigma_i^b \sigma_i^b}_{1} =$$

$$= \frac{1}{2} \sum_{a,b=1}^{m} \left(\sum_{i=1}^{N} \sigma_i^a \sigma_i^b \right)^2 - \frac{N m^2}{2}$$

$$\sum_{a<b} = \frac{1}{2} \left(\sum_{a,b=1}^{m} - \sum_{a,b=1}^{m} \delta_{ab} \right) = \frac{1}{2} \sum_{a,b=1}^{m} (1 - \delta_{ab}) \qquad (*)$$

$$\Rightarrow \quad \frac{1}{2} \sum_{a,b=1}^{m} = \sum_{a<b} + \frac{1}{2} \sum_{a,b=1}^{m} \delta_{ab}$$

$$\sum_{i<j} \sum_{a,b} \sigma_i^a \sigma_j^a \sigma_i^b \sigma_j^b =$$

$$= \sum_{a<b} \left(\sum_{i=1}^{N} \sigma_i^a \sigma_i^b \right)^2 + \frac{1}{2} \sum_{a=1}^{m} \left(\sum_{i=1}^{N} \underbrace{\sigma_i^a \sigma_i^a}_{1} \right)^2 - \frac{N m^2}{2} =$$

$$= \sum_{a<b} \left(\sum_{i} \sigma_i^a \sigma_i^b \right)^2 + \frac{N^2 m - N m^2}{2}$$

Thus

$$\mathbb{E}[Z_J^m] = \sum_{\{\vec{\sigma}\}} e^{\beta h \sum_a \sum_i \sigma_i^a} e^{\frac{\beta^2}{4}(Nm - m^2)} \prod_{a<b} e^{\frac{\beta^2}{2N} \left(\sum_i \sigma_i^a \sigma_i^b \right)^2} =$$

$$= \sum_{\{\vec{\sigma}\}} e^{\beta h \sum_a \sum_i \sigma_i^a} e^{\frac{\beta^2}{4}(Nm - m^2)} \prod_{a<b} e^{\frac{1}{2N\beta^2} \left(\beta^2 \sum_i \sigma_i^a \sigma_i^b \right)}$$

Using again

$$\int dx \, e^{-\frac{1}{2} A x^2 + B x} = \sqrt{\frac{2\pi}{A}} \, e^{\frac{B^2}{2A}}$$

(from right to left, so called Hubbard-Stratonovich transformation)

with

$$A = N\beta^2 \quad , \quad B = \beta^2 \sum_i \sigma_i^a \sigma_i^b$$

Thus

$$\mathbb{E}[Z_J^m] = \sum_{\{\vec{\sigma}\}} e^{\beta h \sum_a \sum_i \sigma_i^a} e^{\frac{\beta^2}{4}(Nm - m^2)} \prod_{a<b} \left(\frac{N\beta^2}{2\pi} \right)^{1/2} \int dQ_{ab} \, e^{-\frac{N\beta^2}{2} Q_{ab}^2 + \beta^2 \sum_i \sigma_i^a \sigma_i^b Q_{ab}}$$

$$\prod_{a<b} \left(\frac{N\beta^2}{2\pi} \right)^{1/2} = \left(\frac{N\beta^2}{2\pi} \right)^{\frac{1}{2} \sum_{a<j} 1}$$

Using (*)

$$\sum_{a<j} 1 = \frac{1}{2} \left(\sum_{a,b=1}^{m} 1 - \sum_{a,b=1}^{m} \delta_{ab} \right) = \frac{m^2 - m}{2} = \frac{m(m-1)}{2}$$

$$\Rightarrow \quad \prod_{a<b} \left(\frac{N\beta^2}{2\pi} \right)^{1/2} = \left(\frac{N\beta^2}{2\pi} \right)^{\frac{m(m-1)}{4}}$$

Thus

$$\mathbb{E}[Z_J^m] = \sum_{\{\vec\sigma\}} e^{\beta h \sum_a \sum_i \sigma_i^a} e^{\frac{\beta^2}{4}(Nm - m^2)} \left(\frac{N\beta^2}{2\pi}\right)^{\frac{m(m-1)}{4}}$$

$$\int \left(\prod_{a<b} dQ_{ab}\right) e^{-\frac{N\beta^2}{2}\sum_{a<b} Q_{ab}^2 + \beta^2 \sum_{a<b}\sum_i \sigma_i^a \sigma_i^b Q_{ab}}$$

Note: in the paper wrong prefactor

$$\left(\frac{2\pi\beta^2}{N}\right)^{\frac{m(m-1)}{2}}$$

We have decoupled the spins inside one replica, but we have introduced a matrix $(Q_{ab})_{a,b=1,\dots,m}$

Hence

$$\mathbb{E}[Z_J^m] = \left(\frac{N\beta^2}{2\pi}\right)^{\frac{m(m-1)}{4}} e^{\frac{\beta^2}{4}(Nm - m^2)}$$

$$\int \left(\prod_{a<b} dQ_{ab}\right) e^{-\frac{N\beta^2}{2}\sum_{a<b} Q_{ab}^2}$$

$$\left(\prod_a \prod_i \sum_{\sigma_i^a = -1,1}\right) \prod_i e^{\beta h \sum_a \sigma_i^a + \beta^2 \sum_{a<b} \sigma_i^a \sigma_i^b Q_{ab}} =$$

$$= \left(\frac{N\beta^2}{2\pi}\right)^{\frac{m(m-1)}{2}} e^{\frac{\beta^2}{4}(Nm - m^2)}$$

$$\int \left(\prod_{a<b} dQ_{ab}\right) e^{-\frac{N\beta^2}{2}\sum_{a<b} Q_{ab}^2}$$

$$\left[\left(\prod_a \sum_{\sigma^a = -1,1}\right) e^{\beta h \sum_a \sigma^a + \beta^2 \sum_{a<b} \sigma^a \sigma^b Q_{ab}}\right]^N$$

Thus

$$\mathbb{E}[Z_J^m] \propto e^{\frac{\beta^2}{4} N m} \int \left(\prod_{a<b} dQ_{ab}\right) e^{-\frac{N\beta^2}{2}\sum_{a<b} Q_{ab}^2}$$

$$e^{N \ln\left[\left(\prod_a \sum_{\sigma^a = -1,1}\right) \exp\left(\beta h \sum_a \sigma^a + \beta^2 \sum_{a<b} \sigma^a \sigma^b Q_{ab}\right)\right]}$$

:

$$= \int \left(\prod_{a<b} dQ_{ab}\right) e^{-N S[Q,h]}$$

where

$$S[Q,h] = -\frac{\beta^2 m}{4} + \frac{\beta^2}{2}\sum_{a<b} Q_{ab}^2 - W[Q,h]$$

with

$$W[Q,h] = \ln\left[\left(\prod_a \sum_{\sigma^a = -1,1}\right) e^{\beta h \sum_a \sigma^a + \beta^2 \sum_{a<b} \sigma^a \sigma^b Q_{ab}}\right]$$

$S[Q, h]$ is called effective action

Saddle-point approximation

$$\mathbb{E}[Z_j^m] = e^{-N S[Q^*, h]}$$

where Q^* maximize $S[Q, h]$

Then

$$f = -\frac{1}{\beta} \lim_{m \to 0} \lim_{N \to \infty} \frac{\ln(\mathbb{E}[Z_j^m])}{N m} =$$

$$= -\frac{1}{\beta} \lim_{m \to 0} \frac{-S[Q^*, h]}{m} =$$

$$= \frac{1}{\beta} \lim_{m \to 0} \frac{S[Q^*, h]}{m}$$

Replica symmetric solution

$S[Q]$ is symmetric under permutation over replicas:

$$S[Q_{ab}] = S[Q_{\pi(a)\pi(b)}]$$

where π is an arbitrary permutation of replica indices.

Ansatz: $(Q^*_{ab}) = (Q^*_{\pi(a)\pi(b)})$

(solution is invariant under the symmetry group of $S[Q]$)

Solution: $Q_{aa} = 0$, $Q_{ab} = q$ $a \neq b$

(replica-symmetric matrix)

Then plugging into $S[Q, h]$ we get

$$S[q, h] = -\frac{\beta^2 m}{4} + \frac{\beta^2}{2} \sum_{a<b} Q_{ab}^2 - \ln\left[\left(\prod_a \sum_{\sigma^a = -1, 1} \right) e^{\beta h \sum_a \sigma^a + \beta^2 \sum_{a<b} \sigma^a \sigma^b Q_{ab}} \right]$$

$$\sum_{a<b} Q_{ab}^2 = q^2 \sum_{a<b} 1 = q^2 \frac{m(m-1)}{2} \overset{m \to 0}{\simeq} -q^2 \frac{m}{2}$$

thus

$$S[q,h] = -\frac{\beta^2 m}{4} - \frac{m\beta^2 q^2}{4} - \ln\left[\left(\prod_a \sum_{\sigma^a=-1,1}\right) e^{\beta h \sum_a \sigma^a} e^{\beta^2 q \sum_{a<b} \sigma^a \sigma^b}\right]$$

$$\sum_{a<b} \sigma^a \sigma^b = \frac{1}{2}\left(\sum_{a,b=1}^{m} - \sum_{a,b=1}^{m} \delta_{ab}\right)\sigma^a\sigma^b =$$

$$= \frac{1}{2}\left(\sum_a \sigma^a\right)^2 - \frac{1}{2}\sum_a \underbrace{\sigma^a\sigma^a}_{1} =$$

$$= \frac{1}{2}\left(\sum_a \sigma^a\right)^2 - \frac{m}{2}$$

Thus

$$S[q,h] =$$

$$= -\frac{\beta^2 m}{4} - \frac{m\beta^2 q^2}{4} - \ln\left[\left(\prod_a \sum_{\sigma^a=-1,1}\right) e^{\beta h \sum_a \sigma^a} e^{-\frac{m\beta^2 q}{2}} e^{\frac{\beta^2 q}{2}\left(\sum_a \sigma^a\right)^2}\right] =$$

$$= -\frac{\beta^2 m}{4} - \frac{m\beta^2 q^2}{4} + \frac{m\beta^2 q}{2} - \ln\left[\left(\prod_a \sum_{\sigma^a=-1,1}\right) e^{\beta h \sum_a \sigma^a} e^{\frac{\beta^2 q}{2}\left(\sum_a \sigma^a\right)^2}\right]$$

Using again

$$\int dx\, e^{-\frac{1}{2}Ax^2 + Bx} = \sqrt{\frac{2\pi}{A}}\, e^{\frac{B^2}{2A}}$$

with

$$A = 1 \quad , \quad B = \beta\sqrt{q}\sum_a \sigma^a$$

then

$$\ln\left[\left(\prod_a \sum_{\sigma^a=-1,1}\right) e^{\beta h \sum_a \sigma^a} e^{\frac{\beta^2 q}{2}\left(\sum_a \sigma^a\right)^2}\right] =$$

$$= \ln\left[\left(\prod_a \sum_{\sigma^a=-1,1}\right) e^{\beta h \sum_a \sigma^a} \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2} + \beta\sqrt{q}\sum_a \sigma^a z}\right] =$$

$$= \ln\left[\int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \left(\prod_a \sum_{\sigma^a=-1,1}\right) e^{(\beta h + \beta\sqrt{q}\, z)\sum_a \sigma^a}\right] =$$

$$= \ln\left[\int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \prod_a \sum_{\sigma^a=-1,1} e^{(\beta h + \beta\sqrt{q}\, z)\sigma^a}\right] =$$

$$= \ln\left[\int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \prod_a 2\cosh(\beta h + \beta\sqrt{q}\, z)\right] =$$

$$= \ln \left\{ \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \left[2 \cosh \left(\beta h + \beta \sqrt{q}\, z \right) \right]^m \right\} =$$

$$= \ln \left\{ \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} e^{m \ln \left[2 \cosh \left(\beta h + \beta \sqrt{q}\, z \right) \right]} \right\} \simeq$$

$$\overset{m \to 0}{\simeq} \ln \left\{ \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \left[1 + m \ln \left[2 \cosh \left(\beta h + \beta \sqrt{q}\, z \right) \right] \right] \right\} =$$

$$= \ln \left\{ 1 + m \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \ln \left[2 \cosh \left(\beta h + \beta \sqrt{q}\, z \right) \right] \right\} \simeq$$

$$\overset{m \to 0}{\simeq} m \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \ln \left[2 \cosh \left(\beta h + \beta \sqrt{q}\, z \right) \right]$$

Therefore

$$S[q, h] = -\frac{\beta^2 m}{4} - \frac{m \beta^2 q^2}{4} + \frac{m \beta^2 q}{2} -$$

$$- m \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \ln \left[2 \cosh \left(\beta h + \beta \sqrt{q}\, z \right) \right] =$$

$$= -m \frac{\beta^2}{4} \left(q^2 - 2q + 1 \right) -$$

$$- m \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \ln \left[2 \cosh \left(\beta h + \beta \sqrt{q}\, z \right) \right] =$$

$$= -m \left\{ \frac{\beta^2}{4} (q-1)^2 + \right.$$

$$\left. + \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \ln \left[2 \cosh \left(\beta h + \beta \sqrt{q}\, z \right) \right] \right\}$$

$$\tilde{f}(q, h) = \frac{1}{\beta} \lim_{m \to 0} \frac{S[q, h]}{m} =$$

Then

$$f = \tilde{f}(q^*, h)$$

where q^* maximize $\tilde{f}(q, h)$ (maximize $S[q, h]$)

We have

$$\tilde{f}(q, h) = -\left\{ \frac{\beta}{4} (1-q)^2 + \right.$$

$$\left. + \frac{1}{\beta} \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \ln \left[2 \cosh \left(\beta h + \beta \sqrt{q}\, z \right) \right] \right\}$$

$$\frac{\partial \tilde{f}(q,h)}{\partial q} = -\left\{ \frac{\beta}{2}(1-q)(-1) + \right.$$
$$\left. + \frac{1}{\beta} \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \frac{z \sinh(\beta h + \beta\sqrt{q}\, z)}{2\cosh(\beta h + \beta\sqrt{q}\, z)} \frac{\beta z}{2} \frac{1}{\sqrt{q}} \right\} =$$

$$= \frac{\beta}{2}(1-q) - \frac{1}{2\sqrt{q}} \int \frac{dz}{\sqrt{2\pi}} z e^{-\frac{z^2}{2}} tgh(\beta h + \beta\sqrt{q}\, z)$$

$$\frac{d}{dz} e^{-\frac{z^2}{2}} = -z e^{-\frac{z^2}{2}}$$

$$\frac{\partial \tilde{f}(q,h)}{\partial q} = \frac{\beta}{2}(1-q) +$$
$$+ \frac{1}{2\sqrt{q}} \int \frac{dz}{\sqrt{2\pi}} \frac{d}{dz}\left(e^{-\frac{z^2}{2}}\right) tgh(\beta h + \beta\sqrt{q}\, z) =$$

$$= \frac{\beta}{2}(1-q) -$$
$$- \frac{1}{2\sqrt{q}} \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \left[1 - tgh^2(\beta h + \beta\sqrt{q}\, z)\right] \beta\sqrt{q} =$$

$$= \frac{\beta}{2}\left\{ 1 - q - 1 + \right.$$
$$\left. + \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} tgh^2(\beta h + \beta\sqrt{q}\, z) \right\} =$$

$$= -\frac{\beta}{2}\left\{ q - \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} tgh^2(\beta h + \beta\sqrt{z}\, z) \right\}$$

$$q^* \quad \left| \quad \frac{\partial \tilde{f}(q,h)}{\partial q}\right|_{q=q^*} = 0$$

$$\Rightarrow \qquad q^* = \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} tgh^2(\beta h + \beta\sqrt{q^*}\, z)$$

(self consistent equation)

$tgh^2(\cdot) \in [0,1]$

$\Rightarrow \quad q^* \in [0,1]$

Therefore

$$f = \tilde{f}(q^*, h)$$

$$= -\frac{\beta}{4}(1-q^*)^2 - \frac{1}{\beta} \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \ln\left[2\cosh(\beta h + \beta\sqrt{q^*}\, z)\right]$$

$h = 0$

Self consistency equation becomes

$$q^* = \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} tgh^2(\beta\sqrt{q^*}\, z)$$

$q^* = 0$ is a solution

In particular, for $T \to +\infty$, $\beta \to 0^+$

$\Rightarrow \quad tgh(\beta \sqrt{q^*}\, z) \longrightarrow 0$

$\Rightarrow \quad q^* = 0$

$$\tilde{f}(q,0) = -\left\{ \frac{\beta}{4}(1-q)^2 + \right.$$
$$\left. + \frac{1}{\beta} \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \ln\left[2\cosh(\beta\sqrt{q}\, z) \right] \right\} =$$

$$= -\left\{ \frac{\beta}{4}(1-q)^2 + \frac{\ln 2}{\beta} + \right.$$
$$\left. + \frac{1}{\beta} \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \ln\left[\cosh(\beta\sqrt{q}\, z) \right] \right\}$$

Expansion around $q = q^* = 0$

$$\ln\left[\cosh(\beta\sqrt{q}\, z)\right] = \frac{1}{2}\beta^2 q z^2 - \frac{1}{12}\beta^4 q^2 z^4 +$$
$$+ O(\beta^6 q^3 z^6)$$

m even

$$\int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} z^m = -\int \frac{dz}{\sqrt{2\pi}} \frac{d}{dz}\left(e^{-\frac{z^2}{2}}\right) z^{m-1} =$$

$$= (m-1) \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} z^{m-2} =$$

$$= \dots =$$

$$= (m-1)(m-3) \dots 1 \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} =$$

$$= (m-1)(m-3) \dots 1$$

Thus

$$\tilde{f}(q,0) = -\left\{ \frac{\beta}{4} - \cancel{\frac{\beta}{2} q} + \frac{\beta}{4} q^2 + \frac{\ln 2}{\beta} + \cancel{\frac{\beta}{2} q} - \right.$$
$$\left. - \frac{\beta^3}{12} q^2 \cdot 3 + O(\beta^5 q^3) \right\}$$

$$= -\left(\frac{\beta}{4} + \frac{\ln 2}{\beta}\right) - \frac{\beta}{4}\left(1-\beta^2\right) q^2 + O(\beta^5 q^3)$$

Quadratic term change sign when

$$1 - \beta^2 = 0 \quad \Rightarrow \quad \beta = 1 \qquad (\beta > 0)$$

Therefore at $h = 0$, $\beta = \beta_c = 1$ we have a phase transition:

- $\beta < \beta_c \quad \Rightarrow \quad q^* = 0$
- $\beta > \beta_c \quad \Rightarrow \quad 0 < q^* \leq 1$ (spin glass phase)

For $h \neq 0$ there is no phase transition.

Moreover, for $h = 0$

$$q^* = \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} tgh^2(\beta \sqrt{q^*} z)$$

$T \to 0^+, \quad \beta \to +\infty$

$$q^* = \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} tgh^2(\beta \sqrt{q^*} z) =$$

$$= \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \left[1 - sech^2(\beta \sqrt{q^*} z)\right] =$$

$$= 1 - \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} sech^2(\beta \sqrt{q^*} z)$$

$$y = \beta \sqrt{q^*} z \quad \Rightarrow \quad z = \frac{y}{\beta \sqrt{q^*}} \to 0$$

$$dz = \frac{dy}{\beta \sqrt{q^*}}$$

then

$$q^* \simeq 1 - \int \frac{dy}{\sqrt{2\pi}\, \beta \sqrt{q^*}} sech^2(y) =$$

$$= 1 - \frac{1}{\sqrt{2\pi}\, \beta \sqrt{q^*}} \left[\tanh(y)\right]_{-\infty}^{+\infty} =$$

$$= 1 - \frac{2}{\sqrt{2\pi}\, \beta \sqrt{q^*}}$$

$$q^* \simeq 1$$

Hence

$$q^* \simeq 1 - \sqrt{\frac{2}{\pi}}\, T \qquad \text{as} \qquad T \to 0^+$$

$$f = -\frac{\beta}{4}(1 - q^*)^2 - \frac{1}{\beta} \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \ln\left[2 \cosh(\beta \sqrt{q^*} z)\right]$$

$$\cosh x = \frac{e^x + e^{-x}}{2} \simeq \frac{e^x}{2} \qquad \text{as} \qquad x \to \infty$$

then

$$f \simeq -\frac{\beta}{4}(1 - q^*)^2 - \frac{1}{\beta} \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} (\beta \sqrt{q^*} z) =$$

$$= -\frac{\beta}{4}(1 - q^*)^2$$

$$s = -\frac{\partial f}{\partial T} = -\frac{\partial f}{\partial \beta}\frac{\partial \beta}{\partial T} = \frac{1}{T^2}\frac{\partial f}{\partial \beta} = \beta^2 \frac{\partial f}{\partial \beta}$$

$$\simeq -\frac{\beta^2}{4}(1-q^*)^2$$

$$q^* \simeq 1 - \sqrt{\frac{2}{\pi}}\frac{1}{\beta}$$

$$\Rightarrow \quad s \simeq -\frac{\beta^2}{4}\left(\sqrt{\frac{2}{\pi}}\frac{1}{\beta}\right)^2 = -\frac{1}{4}\frac{2}{\pi} = -\frac{1}{2\pi}$$

So the entropy at $T=0$ is negative

$$s(0) = -\frac{1}{2\pi} \simeq -0.159 < 0$$

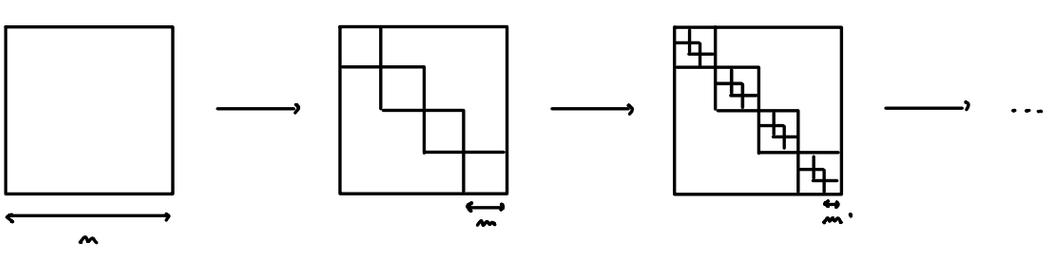
This indicates that the replica symmetric solution is not correct. Furthermore, Looking at the stability of the replica symmetric solution leads to the same conclusion.

From now on we set $h=0$.

Breaking replica symmetry

It is necessary to break the replica symmetry. We will see that breaking the symmetry once (1-RSB) is not sufficient, it will be necessary to break the symmetry an infinite number of times $\longrightarrow$ full-RSB (Parisi solution)

Schematically:



RS 1-RSB 2-RSB

1 - R S B :

The n replicas are divided into $\frac{n}{m}$ groups, each group contains m replicas.

m is an integer number, $1 \leq m \leq n$.

$m = n$ we get back the replica symmetric solution.

Then when we take the limit $n \to 0$, m will be promoted to a real number and $0 \leq m \leq 1$.

If two replicas belong to the same group then the overlap matrix takes value q_1, while if they belong to different groups, it takes value q_0:

$$Q_{ab} = \begin{cases} q_1 & \text{if} \quad I\left(\frac{a}{m}\right) = I\left(\frac{b}{m}\right) \\ q_0 & \text{if} \quad I\left(\frac{a}{m}\right) = I\left(\frac{b}{m}\right) \end{cases}$$

where $I(z)$ is the smallest integer greater or equal to z

E.g.: $I(1.73) = 2$, $I(2) = 2$

E.g.: $n = 12$, $m = 4$

$\frac{n}{m} = 3$

1 2 3 4 | 5 6 7 8 | 9 10 11 12

$I\left(\frac{3}{4}\right) = 1$, $I\left(\frac{4}{4}\right) = 1$ → same group

$I\left(\frac{3}{4}\right) = 1$, $I\left(\frac{11}{4}\right) = 3$ → different groups

Plugging into $S[Q,0]$

$$S[Q,0] = -\frac{\beta^2 m}{4} + \frac{\beta^2}{2} \sum_{a<b} Q_{ab}^2 - \ln\left[\left(\prod_a \sum_{\sigma^a=-1,1}\right) e^{\beta^2 \sum_{a<b} \sigma^a \sigma^b Q_{ab}}\right]$$

$$\sum_{a<b} = \frac{1}{2}\left(\sum_{a,b=1}^{m} - \sum_{a,b=1}^{m} \delta_{ab}\right) =$$

$$= \frac{1}{2}\left[\sum_{\substack{a,b=1 \\ I(\frac{a}{m})=I(\frac{b}{m})}}^{m} + \sum_{\substack{a,b=1 \\ I(\frac{a}{m})\neq I(\frac{b}{m})}}^{m} - \sum_{a,b=1}^{m} \delta_{ab}\right]$$

$$\sum_{\substack{a,b=1 \\ I(\frac{a}{m})=I(\frac{b}{m})}}^{m} 1 = m \cdot m$$

$$\sum_{\substack{a,b=1 \\ I(\frac{a}{m})\neq I(\frac{b}{m})}}^{m} 1 = m \cdot (n-m)$$

$$\sum_{a<b} Q_{ab}^2 = \frac{1}{2}\left[q_1^2 \cdot m \cdot m + q_0^2 \cdot m \cdot (n-m) - q_1^2 m\right] =$$

$$= \frac{1}{2} m \left[q_0^2 (n-m) - q_1^2 (1-m)\right] \simeq$$

$$\overset{m\to 0}{\simeq} \frac{1}{2} m \left[-m q_0^2 - (1-m) q_1^2\right]$$

$$\sum_{a<b} \sigma^a \sigma^b Q_{ab} =$$

$$= \frac{1}{2}\left[q_1 \sum_{\substack{a,b=1 \\ I(\frac{a}{m})=I(\frac{b}{m})}}^{m} \sigma^a \sigma^b + q_0 \sum_{\substack{a,b=1 \\ I(\frac{a}{m})\neq I(\frac{b}{m})}}^{m} \sigma^a \sigma^b - q_1 \sum_{a=1}^{m} \underbrace{\sigma^a \sigma^a}_{1}\right] =$$

$$= \frac{1}{2}\left[q_1 \sum_{\substack{a,b=1 \\ I(\frac{a}{m})=I(\frac{b}{m})}}^{m} \sigma^a \sigma^b + q_0 \sum_{a,b=1}^{m} \sigma^a \sigma^b - q_0 \sum_{\substack{a,b=1 \\ I(\frac{a}{m})=I(\frac{b}{m})}}^{m} \sigma^a \sigma^b - q_1 m\right] =$$

$$= \frac{1}{2}\left[q_0 \left(\sum_{a=1}^{m} \sigma^a\right)^2 + (q_1 - q_0) \sum_{\substack{a,b=1 \\ I(\frac{a}{m})=I(\frac{b}{m})}}^{m} \sigma^a \sigma^b - q_1 m\right]$$

$$\sum_{a,b=1}^{m} \sigma^a \sigma^b = \sum_{K=1}^{m/m} \sum_{a,b=1}^{m} \sigma^{a+m(K-1)} \sigma^{b+m(K-1)} =$$

$I\left(\frac{a}{b}\right) = I\left(\frac{b}{m}\right)$

$$= \sum_{K=1}^{m/m} \left(\sum_{a=1}^{m} \sigma^{a,K} \right)^2$$

where

$$\sigma^{a,K} = \sigma^{a+m(K-1)}$$

thus

$$\sum_{a<b} \sigma^a \sigma^b Q_{ab} =$$

$$= \frac{1}{2} \left[q_0 \left(\sum_{a=1}^{m} \sigma^a \right)^2 + (q_1 - q_0) \sum_{K=1}^{m/m} \left(\sum_{a=1}^{m} \sigma^{a,K} \right)^2 - \right.$$

$$\left. - q_1 m \right]$$

Then

$$S[q_1, q_0, m] =$$

$$= -\frac{\beta^2 m}{4} - \frac{\beta^2 m}{4} \left[m q_0^2 + (1-m) q_1^2 \right] -$$

$$- \ln \left[\left(\prod_a \sum_{\sigma^a = -1,1} \right) e^{\frac{\beta^2}{2} q_0 \left(\sum_{a=1}^{m} \sigma^a \right)^2} \right.$$

$$e^{\frac{\beta^2}{2}(q_1 - q_0) \sum_{K=1}^{m/m} \left(\sum_{a=1}^{m} \sigma^{a,K} \right)^2}$$

$$\left. e^{-\frac{\beta^2}{2} q_1 m} \right] =$$

$$= -\frac{\beta^2 m}{4} - \frac{\beta^2 m}{4} \left[m q_0^2 + (1-m) q_1^2 \right] + \frac{\beta^2 m}{2} q_1 -$$

$$- \ln \left[\left(\prod_a \sum_{\sigma^a = -1,1} \right) e^{\frac{\beta^2}{2} q_0 \left(\sum_{a=1}^{m} \sigma^a \right)^2} \right.$$

$$\left. \prod_{K=1}^{m/m} e^{\frac{\beta^2}{2}(q_1 - q_0) \left(\sum_{a=1}^{m} \sigma^{a,K} \right)^2} \right] =$$

$$= -\frac{\beta^2 m}{4} \left[1 + m q_0^2 + (1-m) q_1^2 - 2 q_1 \right] -$$

$$- \ln \left[\left(\prod_{a=1}^{m} \sum_{\sigma^a = -1,1} \right) e^{\frac{\beta^2}{2} q_0 \left(\sum_{a=1}^{m} \sigma^a \right)^2} \right.$$

$$\left. \prod_{K=1}^{m/m} e^{\frac{\beta^2}{2}(q_1 - q_0) \left(\sum_{a=1}^{m} \sigma^{a,K} \right)^2} \right]$$

Using

$$\int dx\, e^{-\frac{1}{2} A x^2 + B x} = \sqrt{\frac{2\pi}{A}}\, e^{\frac{B^2}{2A}}$$

with

$$A = \frac{1}{q_0} \quad , \quad B = \beta \sum_a \sigma^a$$

then

$$e^{\frac{\beta^2}{2} q_0 \left(\sum_a \sigma^a \right)^2} = \int \frac{dz}{\sqrt{2\pi q_0}}\, e^{-\frac{z^2}{2 q_0}}\, e^{\beta \left(\sum_a \sigma^a \right) z}$$

and with

$$A = \frac{1}{q_1 - q_0} \quad , \quad B = \beta \sum_{a=1}^{m} \sigma^{a,k}$$

then

$$e^{\frac{\beta^2}{2}(q_1-q_0)\left(\sum_{a=1}^{m}\sigma^{a,k}\right)^2} = \int \frac{dy_k}{\sqrt{2\pi(q_1-q_0)}}\, e^{-\frac{y_k^2}{2(q_1-q_0)}}\, e^{\beta\left(\sum_{a=1}^{m}\sigma^{a,k}\right)y_k}$$

Thus

$$\ln\left[\left(\prod_{a}\sum_{\sigma^a=-1,1}\right) e^{\frac{\beta^2}{2}q_0\left(\sum_a \sigma^a\right)^2} \prod_{k=1}^{n/m} e^{\frac{\beta^2}{2}(q_1-q_0)\left(\sum_{a=1}^{m}\sigma^{a,k}\right)^2}\right] =$$

$$= \ln\left[\left(\prod_{k=1}^{n/m}\prod_{a=1}^{m}\sum_{\sigma^{a,k}=-1,1}\right) \int \frac{dz}{\sqrt{2\pi q_0}}\, e^{-\frac{z^2}{2q_0}} \left(\prod_{k=1}^{n/m}\prod_{a=1}^{m} e^{\beta\sigma^{a,k}z}\right) \prod_{k=1}^{n/m}\int \frac{dy_k}{\sqrt{2\pi(q_1-q_0)}}\, e^{-\frac{y_k^2}{2(q_1-q_0)}} \prod_{a=1}^{m} e^{\beta\sigma^{a,k}y_k}\right] =$$

$$= \ln\left\{\int \frac{dz}{\sqrt{2\pi q_0}}\, e^{-\frac{z^2}{2q_0}} \prod_{k=1}^{n/m}\int \frac{dy_k}{\sqrt{2\pi(q_1-q_0)}}\, e^{-\frac{y_k^2}{2(q_1-q_0)}} \prod_{a=1}^{m}\sum_{\sigma^{a,k}=-1,1} e^{\beta(z+y_k)\sigma^{a,k}}\right\} =$$

$$= \ln\left\{\int \frac{dz}{\sqrt{2\pi q_0}}\, e^{-\frac{z^2}{2q_0}} \prod_{k=1}^{n/m}\int \frac{dy_k}{\sqrt{2\pi(q_1-q_0)}}\, e^{-\frac{y_k^2}{2(q_1-q_0)}} \left[2\cosh(\beta(z+y_k))\right]^m\right\} =$$

$$= \ln\left\{\int \frac{dz}{\sqrt{2\pi q_0}}\, e^{-\frac{z^2}{2q_0}} \left[\int \frac{dy}{\sqrt{2\pi(q_1-q_0)}}\, e^{-\frac{y^2}{2(q_1-q_0)}} \left[2\cosh(\beta(z+y))\right]^m\right]^{\frac{n}{m}}\right\} \simeq$$

$$\overset{n\to 0}{\simeq} \ln\left\{\int \frac{dz}{\sqrt{2\pi q_0}}\, e^{-\frac{z^2}{2q_0}} \left[1 + \frac{n}{m}\ln\left[\int \frac{dy}{\sqrt{2\pi(q_1-q_0)}}\, e^{-\frac{y^2}{2(q_1-q_0)}} \left[2\cosh(\beta(z+y))\right]^m\right]\right]\right\} =$$

$$= \ln\left\{1 + \frac{n}{m}\int \frac{dz}{\sqrt{2\pi q_0}}\, e^{-\frac{z^2}{2q_0}} \ln\left[\int \frac{dy}{\sqrt{2\pi(q_1-q_0)}}\, e^{-\frac{y^2}{2(q_1-q_0)}} \left[2\cosh(\beta(z+y))\right]^m\right]\right\} \simeq$$

$$\overset{n\to 0}{\simeq} \frac{n}{m}\int \frac{dz}{\sqrt{2\pi q_0}}\, e^{-\frac{z^2}{2q_0}} \ln\left[\int \frac{dy}{\sqrt{2\pi(q_1-q_0)}}\, e^{-\frac{y^2}{2(q_1-q_0)}} \left[2\cosh(\beta(z+y))\right]^m\right]$$

Therefore, as $m \to 0$

$$S[q_1, q_0, m] =$$

$$= -\frac{\beta^2 m}{4}\left[1 + m q_0^2 + (1-m) q_1^2 - 2 q_1\right] -$$

$$- \frac{m}{m} \int \frac{dz}{\sqrt{2\pi q_0}} e^{-\frac{z^2}{2 q_0}} \ln\left[\int \frac{dy}{\sqrt{2\pi (q_1 - q_0)}} e^{-\frac{y^2}{2(q_1 - q_0)}} \left[2\cosh(\beta(z+y))\right]^m\right]$$

Then

$$\tilde{f}(q_1, q_0, m) = \frac{1}{\beta} \lim_{m \to 0} \frac{S[q_1, q_0, m]}{m} =$$

$$= -\frac{\beta}{4}\left[1 + m q_0^2 + (1-m) q_1^2 - 2 q_1\right] -$$

$$- \frac{1}{m\beta} \int \frac{dz}{\sqrt{2\pi q_0}} e^{-\frac{z^2}{2 q_0}} \ln\left[\int \frac{dy}{\sqrt{2\pi (q_1 - q_0)}} e^{-\frac{y^2}{2(q_1 - q_0)}} \left[2\cosh(\beta(z+y))\right]^m\right]$$

Then we extremize $\tilde{f}(q_1, q_0, m)$ w.r.t. q_1, q_0, m and the free energy reads

$$f = \max_{q_1, q_0, m} \tilde{f}(q_1, q_0, m)$$

Computing the entropy at $T = 0$ we get

$$S_{1\text{-RSB}}(0) \approx -0.01$$

which is still negative, but it is greater than

$$S_{RS}(0) = -\frac{1}{2\pi} \approx -0.159$$

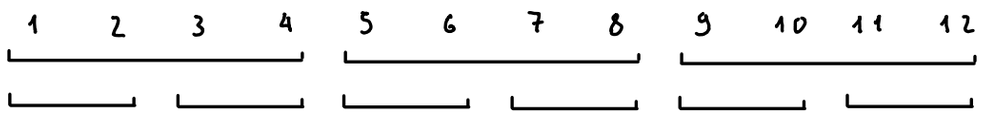
This suggests that we are on the right path.

2-step RSB

We iterate the procedure: each group of m replicas is divided into m'/m subgroups, each subgroup contains m' replicas. $1 \le m' \le m$.

$$Q_{ab} = \begin{cases} q_2 & \text{if } I\left(\frac{a}{m}\right) = I\left(\frac{b}{m}\right),\ I\left(\frac{a}{m'}\right) = I\left(\frac{b}{m'}\right) \\ q_1 & \text{if } I\left(\frac{a}{m}\right) = I\left(\frac{b}{m}\right),\ I\left(\frac{a}{m'}\right) \neq I\left(\frac{a}{m'}\right) \\ q_0 & \text{if } I\left(\frac{a}{m}\right) \neq I\left(\frac{b}{m}\right) \end{cases}$$

E.g. $m = 12$, $m = 4$, $m' = 2$



$I\left(\frac{5}{4}\right) = 2$, $I\left(\frac{7}{4}\right) = 2$ → same group

$I\left(\frac{5}{2}\right) = 3$, $I\left(\frac{7}{2}\right) = 4$ → different subgroup

K-step RSB:

Iterate the procedure K times

$$Q_{ab} = \begin{cases} q_K & \text{if } I\left(\frac{a}{m_i}\right) = I\left(\frac{a}{m_i}\right) \quad i = 1, \dots, K \\ q_{K-1} & \text{if } I\left(\frac{a}{m_i}\right) = I\left(\frac{a}{m_i}\right) \quad i = 1, \dots, K-1 \\ \vdots & \\ q_1 & \text{if } I\left(\frac{a}{m_1}\right) = I\left(\frac{a}{m_1}\right) \\ q_0 & \text{otherwise} \end{cases}$$

with

$$1 \le m_K \le m_{K-1} \le \dots \le m_1 \le m$$

We will get

$$\tilde{f}(q_K, \dots, q_1, q_0, m_K, \dots, m_1)$$

and

$$f = \max \tilde{f}(q_K, \dots, q_1, q_0, m_K, \dots, m_1)$$

The solution gets progressively better, the entropy at $T = 0$ is still negative

$$S_{K\text{-}RSB} < 0$$

but it is closer to zero than the previous steps:

$$S_{K\text{-}RSB} > S_{K-1\text{-}RSB} > \dots > S_{1\text{-}RSB} > S_{RS}$$

This suggests that we have to break the replica symmetry an infinite number of times:

$$K \to \infty$$

$\to$ full RSB

Full RSB:

Define

$$q(x) = q_i \qquad m_i < x < m_{i+1} \quad , \quad i = 0, \dots, K$$

with

$$m_0 = m \quad , \quad m_{K+1} = 1$$

Taking the limit $K \to \infty$, $\tilde{f}$ becomes a functional of $q(x)$:

$$\tilde{f}[q(x)]$$

The free energy is

$$f = \max \tilde{f}[q(x)]$$

To find the maximum

$$\frac{\delta \tilde{f}[q(x)]}{\delta q(x)} = 0$$

Generically the solution can be found numerically, but very close to the critical temperature it is

possible to expand the free energy around $q(x) = 0$

Parisi solution:

$$f = -\frac{\beta}{4}\left[1 + \int_0^1 dx\, q^2(x) - 2q(1)\right] - \frac{1}{\beta}\int \frac{dw}{\sqrt{2\pi}} e^{-\frac{w^2}{2}} f_0(0, \sqrt{q(0)}\, w)$$

where f_0 is the solution to the non-linear antiparabolic equation:

$$\frac{\partial f_0(x,h)}{\partial x} = -\frac{1}{2}\frac{dq(x)}{dx}\left[\frac{\partial^2 f_0(x,h)}{\partial h^2} + x\left(\frac{\partial f_0(x,h)}{\partial h}\right)^2\right]$$

with initial condition:

$$f_0(1,h) = \ln[2\cosh(h)]$$

In this case the entropy at $T = 0$ is 0:

$$S_{\text{full RSB}}(0) = 0$$

Pure states

Summary:

$$f = \frac{1}{\beta}\lim_{m\to 0}\frac{S[Q^*]}{m}$$

where Q^* maximize $S[Q]$.

While the effective action is invariant under permutation of replica indices:

$$S[Q_{ab}] = S[Q_{\pi(a)\pi(b)}]$$

we understood the matrix Q_{ab} cannot be replica symmetric

→ Q_{ab} with a RSB structure

As a consequence there are many different matrices that maximize $S[Q]$.

For instance, if Q_{ab} maximize $S[Q]$, there exists a permutation of replica indices such that

$$Q_{cd} = Q_{\pi(a)\pi(b)} \neq Q_{ab}$$

But we have

$$S[Q_{cd}] = S[Q_{\pi(a)\pi(b)}] = S[Q_{ab}]$$

We call these states pure states, as they stabilize in the thermodynamic limit.
Let us label these pure states with an index α

The Gibbs state is obtained by summing up over all pure states, each weighted by its statistical weight:

$$w_\alpha = \frac{e^{-\beta F_\alpha}}{\sum_\alpha e^{-\beta F_\alpha}}$$

A pure state is a state where the clustering property is satisfied.
Clustering property: all correlations functions must factorize in the long distance limit.

E.g.: $\langle \sigma_i \sigma_j \rangle_\alpha = \langle \sigma_i \rangle_\alpha \langle \sigma_j \rangle_\alpha$

where $\langle \cdot \rangle_\alpha$ is the average on the pure state α.

Consider two different pure states α and β.

Define

$$q_{\alpha\beta} = \frac{1}{N} \sum_{i=1}^{N} \langle \sigma_i \rangle_\alpha \langle \sigma_i \rangle_\beta$$

(overlap between pure states α and β)

We study the statistics of the set $\{ q_{\alpha\beta} \}$.

Introduce

$$P_J(q) = \sum_{\alpha\beta} w_\alpha w_\beta \delta(q_{\alpha\beta} - q)$$

This depend on the specific realization of the random couplings J_{ij}. Then we average over the disorder:

$$P(q) = \mathbb{E}[P_J(q)]$$

→ probability that two pure states have overlap equal to q.

The physical meaning of $q(x)$, introduced in the case of full RSB, is related the probability distribution $P(q)$.

Then we consider $P(q)$ as the order parameter (Paris order parameter).

This manifest the fact that in the full RSB solution we have an

infinite number of order parame-

ters:

$$q_0, q_1, \dots, q_K, \qquad K \to \infty$$

Now, consider the following correla-

tion function:

$$q_J^{(K)} = \frac{1}{N^K} \sum_{i_1,\dots,i_K=1}^{N} \langle \sigma_{i_1} \dots \sigma_{i_K} \rangle^2$$

$$\langle \sigma_{i_1} \dots \sigma_{i_K} \rangle = \sum_\alpha w_\alpha \langle \sigma_{i_1} \dots \sigma_{i_K} \rangle_\alpha =$$

$$= \sum_\alpha w_\alpha \langle \sigma_{i_1} \rangle_\alpha \dots \langle \sigma_{i_K} \rangle_\alpha$$

then

$$q_J^{(K)} = \frac{1}{N^K} \sum_{i_1 \dots i_K=1}^{N} \sum_{\alpha\beta} w_\alpha w_\beta \langle \sigma_{i_1} \rangle_\alpha \langle \sigma_{i_1} \rangle_\beta \dots$$

$$\langle \sigma_{i_K} \rangle_\alpha \langle \sigma_{i_K} \rangle_\beta =$$

$$= \sum_{\alpha\beta} w_\alpha w_\beta \left(\frac{1}{N} \sum_{i_1=1}^{N} \langle \sigma_{i_1} \rangle_\alpha \langle \sigma_{i_1} \rangle_\beta \right) \dots$$

$$\left(\frac{1}{N} \sum_{i_K=1}^{N} \langle \sigma_{i_K} \rangle_\alpha \langle \sigma_{i_K} \rangle_\beta \right) =$$

$$= \sum_{\alpha\beta} w_\alpha w_\beta q_{\alpha\beta}^K =$$

$$= \int dq \sum_{\alpha\beta} w_\alpha w_\beta \delta(q_{\alpha\beta} - q) q^K =$$

$$= \int dq \, P_J(q) \, q^K$$

Then, averaging over the disorder

$$q^{(K)} = \mathbb{E}[q_J^{(K)}] =$$

$$= \int dq \, \mathbb{E}[P_J(q)] \, q^K =$$

$$= \int dq \, P(q) \, q^K$$

In the replica approach

$$q^{(K)} = \mathbb{E}[q_J^{(K)}] =$$

$$= \lim_{m \to 0} \left[\mathbb{E}[\langle \sigma_i^a \sigma_i^b \rangle] \right]^K$$

Identifying

$$Q_{ab} = \mathbb{E}[\langle \sigma_i^a \sigma_i^b \rangle]$$

then

$$q^{(K)} = \lim_{m \to 0} (Q_{ab})^K$$

As we already said there can be many matrices Q_{ab}, then we have to sum over all these.

All the matrices can be obtained starting from one RSB matrix and taking all possible permutations of rows and columns.

The summation over all these possi. ble permutations is equivalent to summing over all possible replica indices:

$$q^{(k)} = \lim_{n \to 0} \frac{1}{n(n-1)} \sum_{\substack{a,b=1 \\ a \neq b}}^{n} (Q_{ab})^k =$$

$$= \int dq \lim_{n \to 0} \frac{1}{n(n-1)} \sum_{\substack{a,b=1 \\ a \neq b}}^{n} \delta(Q_{ab} - q)\, q^k$$

then we identify

$$P(q) = \lim_{n \to 0} \frac{1}{n(n-1)} \sum_{\substack{a,b=1 \\ a \neq b}}^{n} \delta(Q_{ab} - q) =$$

$$= \int_0^1 dx\ \delta(q(x) - q)$$

In the case of the full-RSB solution $q(x)$ is monotonous, then invertible:

$$x(q)$$

From the previous equation

$$P(q) = \frac{dx(q)}{dq}$$

From which we get

$$x(q) = \int_0^q dq'\, P(q')$$

→ $x(q)$, the inverse of $q(x)$, gives the probability of having a

pair of pure states wit overlap between 0 and q.

Fluctuations of $P_J(q)$

$$P_J(q) = \sum_{\alpha\beta} w_\alpha w_\beta \, \delta(q_{\alpha\beta} - q)$$

In the thermodynamic limit $N \to \infty$, $P_J(q)$ approaches a definite limit or fluctuates?

We will see that in the thermodynamic limit, $P_J(q)$ still depend on the realization of the couplings J_{ij}

$\Rightarrow$ $P_J(q)$ is non self-averaging

We want to compute

$$\mathbb{E}\left[P_J(q_1) P_J(q_2)\right] - P_J(q_1) P_J(q_2)$$

Introduce Laplace transform:

$$g(y) = \int dq \, P(q) e^{yq} = \int dq \, \mathbb{E}[P(q)] e^{yq}$$

Using

$$P(q) = \int_0^1 dx \, \delta(q(x) - q)$$

then

$$g(y) = \int dq \int_0^1 dx \, \delta(q(x) - q) e^{yq} =$$
$$= \int_0^1 dx \, e^{y q(x)}$$

To compute $\mathbb{E}[P_J(q_1) P_J(q_2)]$, generalized Laplace transform

$$g(y_1, y_2) = \int dq_1 dq_2 \, P(q_1, q_2) e^{y_1 q_1 + y_2 q_2} =$$
$$= \int dq_1 dq_2 \, \mathbb{E}[P_J(q_1) P_J(q_2)] e^{y_1 q_1 + y_2 q_2}$$

$P(q_1, q_2)$ is the probability that, considering 4 pure state $\alpha_1, \alpha_2, \alpha_3$ and α_4, we have

$$q_{\alpha_1 \alpha_2} = q_1 \quad , \quad q_{\alpha_3 \alpha_4} = q_2$$

Within the replica approach

$$g(y_1, y_2) = \lim_{m \to 0} \frac{1}{m(m-1)(m-2)(m-3)} \sum_{\substack{a,b,c,d=1 \\ a \neq b \neq c \neq d}}^{m} e^{y_1 Q_{ab} + y_2 Q_{cd}}$$

Using the full RSB parametrization, one can show that:

$$\mathbb{E}\left[P_J(q_1) P_J(q_2) \right] = \frac{1}{3} P(q_1) \delta(q_1 - q_2) + \frac{2}{3} P(q_1) P(q_2)$$

Stochastic stability

$\rho(F)$: distribution of the free energies

$\rho(F)\, dF$ is the number of values of F in the interval $[F, F + dF]$

Principle of stochastic stability: the distribution of the free energies, related to the various pure states, is stable under indipendent random increments.
Considering random perturbations not correlated with the original Hamiltonian, we require stochastic stability. Namely these random perturbations change the free energies, but the distribution of the free energies does not change

Consider the simplest case: 1-RSB

$$Q_{ab} = \begin{cases} q_1 & \text{if } I\left(\frac{a}{m}\right) = I\left(\frac{b}{m}\right) \\ q_0 & \text{otherwise} \end{cases}$$

Let's pertube the free energy of a pure state α:

$$F_\alpha' = F_\alpha + \varepsilon r_\alpha$$

r_α: i.i.d. random variables.

$$\rho(F') = \rho(F + \varepsilon r) = \rho(F) + \varepsilon \frac{d\rho}{dF} r + \frac{1}{2}\varepsilon^2 \frac{d^2\rho}{dF^2} r^2 + \\ + O(\varepsilon^2)$$

Stochastic stability requires

$$\rho(F') = \rho(F)$$

hence

$$\frac{d\rho}{dF} \propto \frac{d^2\rho}{dF^2}$$

therefore

$$\rho(F) \propto e^{\beta m F}$$

In order to have

$$\sum_\alpha \rho(F_\alpha) \propto \sum_\alpha e^{\beta m F_\alpha}$$

convergent, it necessary

$$m < 1$$

Stochastic stability fixes the form of $\rho(F)$.

Ultrametricity

We would like to understand the topology of the space of pure states. P(q) does not give enough informations to understand this structure.

Consider three pure states α_1, α_2 and α_3. We compute $P(q_1, q_2, q_3)$, that is the probability of having overlaps

$$q_{\alpha_1\alpha_2} = q_3 \quad , \quad q_{\alpha_2\alpha_3} = q_2 \quad , \quad q_{\alpha_3\alpha_1} = q_1$$

Using Laplace transform

$$g(y_1, y_2, y_3) = \int dq_1 dq_2 dq_3 \; P(q_1, q_2, q_3) \; e^{y_1 q_1 + y_2 q_2 + y_3 q_3}$$

and within the replica approach

$$g(y_1, y_2, y_3) = \lim_{n \to 0} \frac{1}{n(n-1)(n-2)} \sum_{\substack{a,b,c=1 \\ a \neq b \neq c}}^{n} e^{y_1 Q_{ab} + y_2 Q_{bc} + y_3 Q_{ca}}$$

One obtains

$$\begin{aligned} P(q_1, q_2, q_3) = & \frac{1}{2} P(q_1) \chi(q_1) \delta(q_1 - q_2) \delta(q_1 - q_3) + \\ & + \frac{1}{2} P(q_1) P(q_2) \theta(q_1 - q_2) \delta(q_2 - q_3) + \\ & + \frac{1}{2} P(q_2) P(q_3) \theta(q_2 - q_3) \delta(q_3 - q_1) + \\ & + \frac{1}{2} P(q_3) P(q_1) \theta(q_3 - q_1) \delta(q_1 - q_2) \end{aligned}$$

Then $P(q_1, q_2, q_3)$ is non-zero only in the three cases:

$$q_1 = q_2 \leq q_3$$

$$q_1 = q_3 \leq q_2$$

$$q_2 = q_3 \leq q_1$$

Namely, $P(q_1, q_2, q_3)$ is non-zero if all the three parameter coincide or two parameters are equal and the remaining one is larger.

If the overlap between two pure states defines a notion of distance in the space of pure states, the previous property means that in the space of pure states there are only equilateral and isoscel triangles, no scalen triangle.
This property is called ultrame-
tricity.

It was proved by Panchenko that, in general, stochastic stability implies ultrametricity.